

Sujan Parajuli

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Research Profile: Mathematics and Computer Science undergraduate interested in machine learning, statistics, mathematical modeling, and computational finance. Research experience spans graph theory, large-scale healthcare prediction, agricultural data analysis, and ongoing mathematical and machine-learning studies of the Nepal Stock Exchange. Preparing for Ph.D. study involving data-driven mathematical research.

Education

Norwich University, Northfield, VT

Expected May 2027

B.S. in Computer Science and Mathematics, Dual Major

GPA: 3.90/4.00

Minors: Actuarial Science, Information Assurance

Relevant Coursework: Data Structures, Algorithms, Machine Learning, Database Systems, Computer Architecture, Probability, Statistics, Regression Analysis, Linear Algebra, Discrete Mathematics, Calculus I–III

Research and Technical Skills

Programming and Reproducibility: Python, C++, R, SQL, LaTeX, Git, GitHub, Jupyter Notebook, Linux/Unix

Statistical and Machine Learning Methods: Regression, classification, cross-validation, feature engineering, time-series analysis, statistical modeling, model evaluation

Libraries and Data Tools: Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, XGBoost, Excel

Systems: MSP430 Assembly, embedded C, Verilog, computer architecture, Wireshark

Research Experience

Mathematics Research

Mathematics Capstone Researcher – NEPSE Mathematical Modeling

Aug 2026 – Present

Norwich University, Northfield, VT — Mentor: Dr. Sean Kramer

- Will investigate Nepal Stock Exchange price behavior through time-series analysis, return distributions, volatility measures, correlation, and regression to establish a rigorous mathematical foundation for forecasting.
- Will build and evaluate quantitative baselines in Python and Jupyter Notebook, using historical market data to identify patterns and produce interpretable results for the mathematics capstone.
- Will design the mathematical phase to support a subsequent machine-learning study, creating a unified research pipeline connecting statistical assumptions, model performance, and financial interpretation.

Weintz Undergraduate Research Fellow – Vertex Magic Trees

May 2025 – Jul 2025

Norwich University, Northfield, VT — Mentor: Dr. Dan McQuillan

- Investigated vertex-magic total labelings using graph theory, combinatorial arguments, and computational searches, producing new examples and testing conjectured structural bounds.
- Developed algorithms to construct and verify labelings across multiple tree families, using computational experiments to detect patterns and guide proofs.
- Synthesized more than 20 scholarly sources and presented the resulting mathematical arguments, counterexamples, and computational evidence in a research poster and faculty presentation.

Computer Science and Data Research

Computer Science Capstone Researcher – NEPSE Forecasting

Aug 2026 – Present

Norwich University, Northfield, VT — Mentor: Prof. Charles Snow

- Will develop an end-to-end forecasting pipeline for Nepal Stock Exchange data, progressing from regression and classification baselines to sequence-based machine-learning models.
- Will engineer lagged prices, returns, moving averages, volatility, and technical indicators in Python, then compare models through time-aware validation and reproducible Jupyter notebooks.
- Will maintain code, experiments, and documentation with Git and GitHub to create a reproducible capstone suitable for continued research and public technical review.

Undergraduate Researcher – Healthcare Predictive Modeling

Jan 2025 – Apr 2025

Norwich University, Northfield, VT — Mentor: Dr. Natalie Cartwright

- Analyzed more than 30 million HCUP NEDS 2016 emergency-department records to quantify how demographic, insurance, geographic, and socioeconomic factors related to hospital charges.
- Built reproducible Python workflows for cleaning, encoding, outlier treatment, feature preparation, and model training with Pandas, NumPy, Scikit-learn, and XGBoost.
- Compared models with cross-validation and statistical metrics, then communicated findings through faculty discussions, technical documentation, and a research poster while protecting restricted patient-level data.

Teaching Experience

Peer Tutor – Mathematics and Statistics

Aug 2023 – Present

Norwich University, Northfield, VT

- Provided one-on-one tutoring to more than 20 students in calculus, statistics, and pre-calculus, using guided practice and individualized explanations to strengthen independent problem solving.
- Staffed open tutoring hours for two professors across Calculus I, Calculus II, and Elementary Statistics, helping students with course concepts, assignments, and exam preparation.

Professional Experience

Arena IT Helper

2026 – Present

Kreitzberg Arena, Norwich University, Northfield, VT

- Designed more than five digital advertisements for arena programs and hockey promotions, producing brand-consistent materials that supported outreach and season-pass campaigns.
- Supported hockey ticket and season-pass sales through Purplepass while assisting with event scheduling, content updates, and day-to-day technology needs.

Conference Services Team Leader

Summer 2026

Norwich University, Northfield, VT

- Coordinated housing, access, facilities, parking, and event transitions for more than 300 students across multiple summer programs, adapting plans to changing schedules and participant counts.
- Connected camp directors with Public Safety, Facilities, Athletics, custodial teams, and dining staff, resolving time-sensitive operational issues and maintaining accurate room and event information.

Farm Data and IT Assistant

Summer 2025, Summer 2026

Green Mountain Girls Farm, Northfield, VT

- Collected, cleaned, and organized field-level soil-test data across two summer seasons, creating analysis-ready records for comparing nutrient conditions across farm areas.
- Applied statistical and machine-learning methods to soil macronutrient data, identifying spatial patterns that supported evidence-based farm-management decisions.
- Maintained website content and resolved day-to-day technology issues, combining data analysis with operational IT support for a working farm business.

Student Supervisor

Sep 2023 – May 2025

Sodexo, Norwich University, Northfield, VT

- Trained more than five student employees, assigned daily work, and coordinated shift coverage during high-volume service periods to maintain consistent operations.

Leadership and Activities

Vice President, Billiards Club

Jan 2025 – Present

Norwich University, Northfield, VT

- Organize practices, tournaments, scheduling, and member communication for a 20-player club, supporting sustained participation and competitive success.

President, Nepali Student Union

Aug 2024 – Present

Norwich University, Northfield, VT

- Lead the executive board and coordinate cultural events, budgets, logistics, and campus partnerships, expanding opportunities for community engagement and cross-cultural learning.

Orientation Leader

Fall 2025

Norwich University, Northfield, VT

- Guided a group of 43 incoming students through orientation activities, campus resources, schedules, and community expectations, helping them transition successfully to university life.

Junior Civilian Ring Committee Member

2025

Norwich University, Northfield, VT

- Coordinated ceremony logistics with committee members, vendors, and university staff, helping deliver the Junior Ring event for more than 200 attendees.

Certifications and Languages

Certifications: Microsoft and LinkedIn Career Essentials in Data Analysis and Cybersecurity

Languages: English (Fluent), Nepali (Native), Hindi (Conversational)